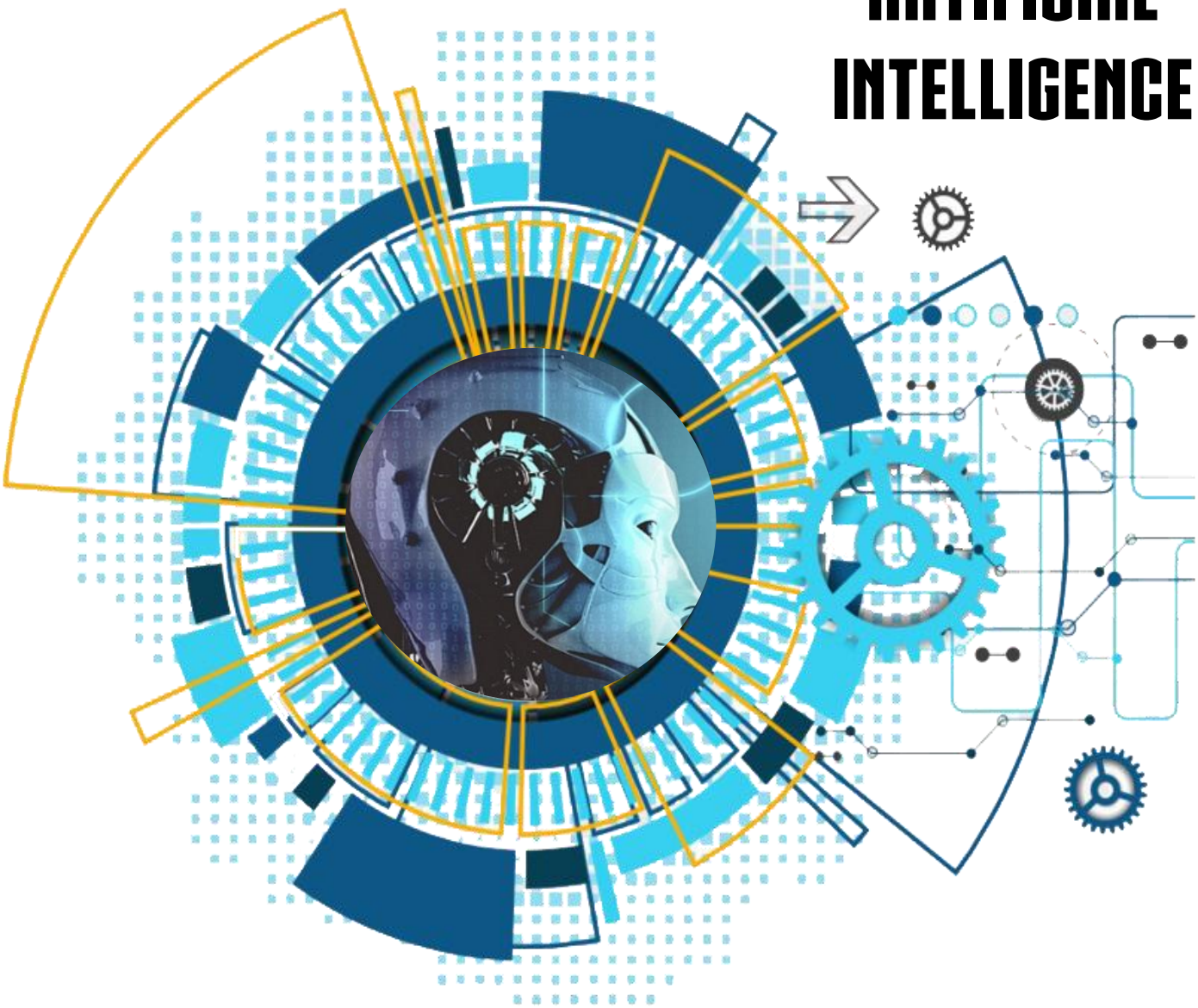
ARTIFICIAL INTELLIGENCE



LEARNING COMPETENCIES

1.1 Understand Artificial Intelligence concepts, its applications, principles, and components, .

1.2 Explain why Artificial Intelligence is important and how it is changing our lives.

INTRODUCTION

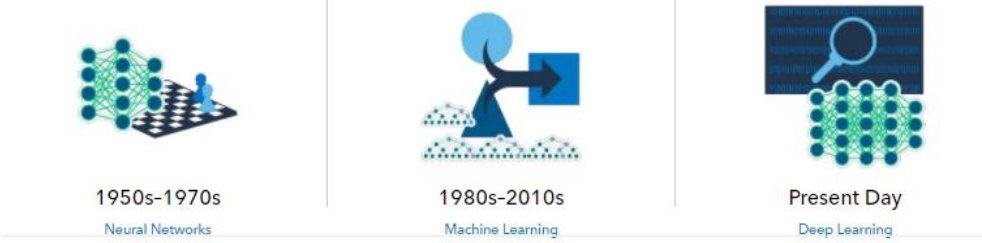
Artificial intelligence (AI) is a wide-ranging branch of computer science concerned with building smart machines capable of performing tasks that typically require human intelligence. It is the study of ideas that enable computers to do the things that make people seem intelligent. The central principles of AI include such as reasoning, knowledge, planning, learning, communication, perception, and the ability to move and manipulate objects. It is the science and engineering of making intelligent machines, especially intelligent computer programs. Many of these artificial intelligence systems are powered by machine learning and deep learning

In this module, we will have the opportunity to explore more about the following topics: history of artificial intelligence, why is artificial intelligence important, how Artificial Intelligence works, how Artificial Intelligence is being used, and advantages, limitations, and applications of Artificial Intelligence

DATA INPUT

Artificial Intelligence History

The term 'Artificial intelligence' was coined by **John McCarthy** in the year **1956** at Dartmouth College at the first-ever AI conference. Later that year, JC Shaw, Herbert Simon, and Allen Newell created the first AI software program named 'Logic Theorist.'
Although, the idea of a 'machine that thinks' dates back to the Mayan civilization. In the modern era, there have been some important events since the advent of electronic computers that played a crucial role in the evolution of Artificial Intelligence:



- **Maturation of Artificial Intelligence (1943–1952):** Walter Pitts and Warren S McCulloch, two mathematicians, published 'A Logical Calculus of the Ideas Immanent in Nervous Activity' in the Journal of Mathematical Biophysics. They described the behavior of human neurons with the help of simple logical functions that inspired an English mathematician Alan Turing to publish 'Computing Machinery and Intelligence' that comprised a test. This **Turing Test** is used to check a machine's ability to exhibit intelligent behavior.

- **The birth of Artificial Intelligence (1952–1956):** Logic Theorist, the first AI program was created in the year 1955 by Allen Newell and Herbert A Simon. It proved around 52 mathematical theorems and improved the proofs for other theorems. Professor John McCarthy coined the term 'Artificial Intelligence' at the Dartmouth conference, and it was accepted as an academic field.

- **Golden years – early enthusiasm (1956–1974):** After the invention of high-level languages such as LISP, COBOL, and FORTRAN, researchers got more excited about AI and developed algorithms to solve complex mathematical problems. Joseph Weizenbaum, a computer scientist, created the first chatbot named 'ELIZA' in the year 1966. A year later, Frank Rosenblatt built a computer named 'Mark 1 Perceptron.' This computer was based on the biological neural network (BNN) and learned through the method of trial and error that was later coined as reinforced learning. In 1972, Japan built the first intelligent humanoid robot named 'WABOT-1.' Since then, robots are constantly being developed and trained to perform complex tasks in various industries.

- **A boom in AI (1980–1987)**: The first AI winter (1974–1980) was over, and governments started seeing the potential of how useful AI systems could be for the economy and defense forces. Expert systems and software were programmed to simulate the decision-making ability of the human brain in machines. Al algorithms like backpropagation, which uses neural networks to understand a problem and find the best possible solution, were used.

- **The AI Winter (1987–1993):** By the end of the year 1988, IBM successfully translated a set of bilingual sentences from English to French. More advancements were going on in the field of AI and Machine Learning, and by 1989, Yann LeCun successfully applied the **backpropagation algorithm** to recognize handwritten ZIP codes. It took three days for the system to produce the results but was still fast enough given the hardware limitations at that time.

- **The emergence of intelligent agents (1993–2011):** In the year 1997, IBM developed a chess-playing computer named 'Deep Blue' that outperformed the world chess champion, Garry Kasparov, in a chess match, twice. In 2002, Artificial intelligence for the first time stepped into the domestics and built a vacuum cleaner named 'Roomba.' By the year 2006, MNCs such as Facebook, Google, and Microsoft started using AI algorithms and Data Analytics to understand customer behavior and improve their recommendation systems.

- **Deep Learning, Big Data, and Artificial General Intelligence (2011–Present):** With computing systems becoming more and more powerful, it is now possible to process large amounts of data and train our machines to make better decisions. Supercomputers take the advantage of AI algorithms and **neural networks** to solve some of the most complex problems of the modern world. Recently, Neuralink, a company owned by Elon Musk, successfully demonstrated a brain–

machine interface where a monkey played the ping pong ball video game from his mind

Why is artificial intelligence important?

- **AI automates repetitive learning and discovery through data.** Instead of automating manual tasks, AI performs frequent, high-volume, computerized tasks. And it does so reliably and without fatigue. Of course, humans are still essential to set up the system and ask the right questions.
- **AI adds intelligence** to existing products. Many products you already use will be improved with AI capabilities, much like Siri was added as a feature to a new generation of Apple products. Automation, conversational platforms, bots and smart machines can be combined with large amounts of data to improve many technologies. Upgrades at home and in the workplace, range from security intelligence and smart cams to investment analysis.
- **AI adapts through progressive learning algorithms** to let the data do the programming. AI finds structure and regularities in data so that algorithms can acquire skills. Just as an algorithm can teach itself to play chess, it can teach itself what product to recommend next online. And the models adapt when given new data.
- **AI analyzes more and deeper data** using neural networks that have many hidden layers. Building a fraud detection system with five hidden layers used to be impossible. All that has changed with incredible computer power and big data. You need lots of data to train deep learning models because they learn directly from the data.
- **AI achieves incredible accuracy** through deep neural networks. For example, your interactions with Alexa and Google are all based on deep learning. And these products keep getting more accurate the more you use them. In the medical field, AI techniques from deep learning and object recognition can now be used to pinpoint cancer on medical images with improved accuracy.
- **AI gets the most out of data.** When algorithms are self-learning, the data itself is an asset. The answers are in the data. You just have to apply AI to find them. Since the role of the data is now more important than ever, it can create a competitive advantage. If you have the best data in a competitive industry, even if everyone is applying similar techniques, the best data will win.

How Artificial Intelligence Works

AI works by combining large amounts of data with fast, iterative processing and intelligent algorithms, allowing the software to learn automatically from patterns or features in the data. AI is a broad field of study that includes many theories, methods and technologies, as well as the following major subfields:

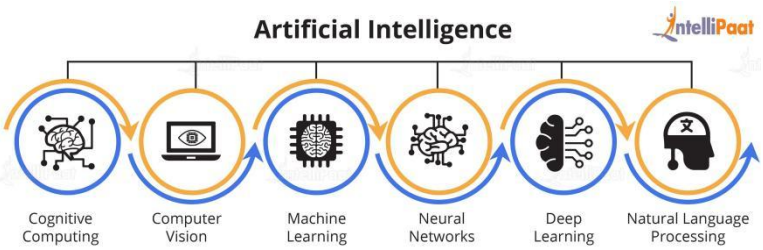


Image source: https://intellipaat.com/blog/what-is-artificial-intelligence/#no1

- **Machine learning** automates analytical model building. It uses methods from neural networks, statistics, operations research and physics to find hidden insights in data without explicitly being programmed for where to look or what to conclude.
- **A neural network** is a type of machine learning that is made up of interconnected units (like neurons) that processes information by responding to external inputs, relaying information between each unit. The process requires multiple passes at the data to find connections and derive meaning from undefined data.
- **Deep learning** uses huge neural networks with many layers of processing units, taking advantage of advances in computing power and improved training techniques to learn complex patterns in large amounts of data. Common applications include image and speech recognition.
- **Cognitive Computing.** The ultimate goal of cognitive computing is to imitate the human thought process in a computer model. How can this be achieved? Using self-learning algorithms, pattern recognition by neural networks, and natural language processing, a computer can mimic the human way of thinking. Here, computerized models are deployed to simulate the human cognition process.
- **Computer vision** relies on pattern recognition and deep learning to recognize what's in a picture or video. When machines can process, analyze and understand images, they can capture images or videos in real time and interpret their surroundings.
- **Natural language processing** (NLP) is the ability of computers to analyze, understand and generate human language, including speech. The next stage of NLP is natural language interaction, which allows humans to communicate with computers using normal, everyday language to perform tasks.

Additionally, several technologies enable and support AI:

- **Graphical processing units** are key to AI because they provide the heavy compute power that's required for iterative processing. Training neural networks requires big data plus compute power.
- **The Internet of Things** generates massive amounts of data from connected devices, most of it unanalyzed. Automating models with AI will allow us to use more of it.
- **Advanced algorithms** are being developed and combined in new ways to analyze more data faster and at multiple levels. This intelligent processing is key to identifying and predicting rare events, understanding complex systems and optimizing unique scenarios.

- **APIs, or application programming interfaces,** are portable packages of code that make it possible to add AI functionality to existing products and software packages. They can add image recognition capabilities to home security systems and Q&A capabilities that describe data, create captions and headlines, or call out interesting patterns and insights in data.

Advantages of Artificial Intelligence

- **Reduced human error:** With humans involved in the tasks where precision is required, there will always be a chance of error. However, if programmed properly, machines do not make mistakes and easily perform repetitive tasks without making many errors, if not at all.

- **Risk avoidance:** Replacing humans with intelligent robots is one of the biggest advantages of Artificial Intelligence. AI robots are now doing risky things replacing humans in places such as coal mines, exploring the deepest parts of the ocean, sewage treatment, and nuclear power plants to avoid any disaster.

- **Replacing repetitive jobs:** Our day-to-day work includes many repetitive tasks that we have to do every day without any change. For example, washing your clothes or mopping the floor doesn't require you to be creative and find new easy to do it every day. Even big industries have production lines where the same number of tasks has to be done in an exact sequence. Now, machines have replaced these tasks so that humans can spend this time doing creative things.

- **Digital assistance:** With digital assistants to interact with users 24/7, organizations can save the need for human resources and deliver faster service to customers. It is a win-win situation for both the organization and the customers. In most cases, it is really hard to determine whether a customer is chatting with the chatbot or a human being.

Limitations of Artificial Intelligence

- **High cost of creation:** It may sound a little spooky, but the rate at which computational devices are upgraded is phenomenal. Machines need to be repaired and maintained with time to keep the latest requirements in check, which needs a lot of resources.

- **No emotions:** There is no doubt that machines are much powerful and faster than human beings. They can perform multiple tasks simultaneously and produce results in a split second. AI-powered robots can also lift more weight, thereby increasing the production cycle. However, machines cannot build an emotional connection with other human beings, which is a crucial aspect of team management.

- **Box thinking:** Machines can perfectly execute the preassigned tasks or operations with a definite range of constraints. However, they start producing ambiguous results if they get anything out of the trend.

- **Can't think for Itself:** Artificial Intelligence aims to process data and make a conscious decision as we humans do. But, at present, it can only do the tasks it is programmed for. AI systems cannot make decisions based on emotions, compassion, and empathy. For example, if a self-driving car is not programmed to consider animals like deer as living organism, it will not stop even if it hits a deer and knock it off.

Applications of AI

Fraud Detection

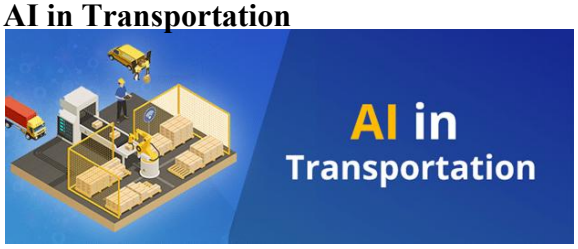
Music and Movie Recommendations

AI in Retail

Autopilot Flight

AI in Healthcare

AI in Transportation



How Artificial Intelligence Is Being Used

Every industry has a high demand for AI capabilities – including systems that can be used for automation, learning, legal assistance, risk notification and research. Specific uses of AI in industry include:

- **Health Care**
  AI applications can provide personalized medicine and X-ray readings. Personal health care assistants can act as life coaches, reminding you to take your pills, exercise or eat healthier.

- **Retail**

AI provides virtual shopping capabilities that offer personalized recommendations and discuss purchase options with the consumer. Stock management and site layout technologies will also be improved with AI.

- **Manufacturing**
  AI can analyze factory IoT data as it streams from connected equipment to forecast expected load and demand using recurrent networks, a specific type of deep learning network used with sequence data.

- **Life sciences**
  From ensuring drug safety to getting new therapies to market faster, AI technologies can unleash the full potential of data to solve some of our greatest health challenges.

- **Banking**
  Artificial Intelligence enhances the speed, precision and effectiveness of human efforts. In financial institutions, AI techniques can be used to identify which transactions are likely to be fraudulent, adopt fast and accurate credit scoring, as well as automate manually intense data management tasks.

- **Public sector**
  Artificial Intelligence can make smart cities smarter. It can support national defense with mission readiness and predictive maintenance. Across the board, AI can improve program efficiency and effectiveness.

- **Game playing.** You can buy machines that can play master level chess for a few hundred dollars. There is some AI in them, but they play well against people mainly through brute force computation—looking at hundreds of thousands of positions. To beat a world champion by brute force and known reliable heuristics requires being able to look at 200 million positions per second.

- **Speech recognition.** In the 1990s, computer speech recognition reached a practical level for limited purposes. Thus United Airlines has replaced its keyboard tree for flight information by a system using speech recognition of flight numbers and city names. It is quite convenient. On the the other hand, while it is possible to instruct some computers using speech, most users have gone back to the keyboard and the mouse as still more convenient.

- **Understanding natural language.** Just getting a sequence of words into a computer is not enough. Parsing sentences is not enough either. The computer has to be provided with an understanding of the domain the text is about, and this is presently possible only for very limited domains.

- **Computer Vision.** The world is composed of three-dimensional objects, but the inputs to the human eye and computers' TV cameras are two dimensional. Some useful programs can work solely in two dimensions, but full computer vision requires partial three-dimensional information that is not just a set of two-dimensional views. At present there

are only limited ways of representing three-dimensional information directly, and they are not as good as what humans evidently use.

- **Expert Systems.** A "knowledge engineer" interviews experts in a certain domain and tries to embody their knowledge in a computer program for carrying out some task. How well this works depends on whether the intellectual mechanisms required for the task are within the present state of AI. When this turned out not to be so, there were many disappointing results. One of the first expert systems was MYCIN in 1974, which diagnosed bacterial infections of the blood and suggested treatments. It did better than medical students or practicing doctors, provided its limitations were observed. Namely, its ontology included bacteria, symptoms, and treatments and did not include patients, doctors, hospitals, death, recovery, and events occurring in time. Its interactions depended on a single patient being considered. Since the experts consulted by the knowledge engineers knew about patients, doctors, death, recovery, etc., it is clear that the knowledge engineers forced what the experts told them into a predetermined framework. In the present state of AI, this has to be true. The usefulness of current expert systems depends on their users having common sense.

- **Heuristic Classification.** One of the most feasible kinds of expert system given the present knowledge of AI is to put some information in one of a fixed set of categories using several sources of information. An example is advising whether to accept a proposed credit card purchase. Information is available about the owner of the credit card, his record of payment and also about the item he is buying and about the establishment from which he is buying it.